

Sixth-order log-determinant convexity for two-dimensional split covariance intersection

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Abstract

Let A, B, C, D be real symmetric positive semidefinite 2×2 matrices, with $A + B$ and $C + D$ positive definite. For $0 < w < 1$, set $U = A/w + B$, $V = C/(1 - w) + D$, and $P = (U^{-1} + V^{-1})^{-1}$. We prove that the sixth derivative of $\log \det P$ with respect to w is nonnegative, keeping the four matrices fixed. This establishes the two-dimensional, sixth-order case of Li's log-determinant conjecture for split covariance intersection. The proof reduces the determinant to a ratio of scalar polynomials. For positive definite inputs, ordered eigenvalue crossings locate and compare all their roots, including multiplicities; reciprocal sixth powers then give the derivative sign. A finite Taylor-coefficient continuity argument covers all admissible singular inputs. No simultaneous diagonalization is assumed.

1 Statement and context

For real symmetric matrices, $M \succeq 0$ and $M \succ 0$ denote positive semidefiniteness and positive definiteness, respectively. Matrix inequalities below use this order, I_2 is the identity matrix, and $\log$ is the natural logarithm.

Theorem 1. *Let $A, B, C, D \in \mathbb{R}^{2 \times 2}$ be symmetric and satisfy*

$$A, B, C, D \succeq 0, \quad A + B \succ 0, \quad C + D \succ 0. \quad (1)$$

For every $w \in (0, 1)$, define

$$U(w) = \frac{A}{w} + B, \quad V(w) = \frac{C}{1 - w} + D, \quad P(w) = (U(w)^{-1} + V(w)^{-1})^{-1}. \quad (2)$$

Then, with A, B, C, D fixed under differentiation,

$$\frac{d^6}{dw^6} \log \det P(w) \geq 0. \quad (3)$$

The correspondence with the split covariance intersection notation is $A = P_{1d}$, $B = P_{1i}$, $C = P_{2d}$, and $D = P_{2i}$. Li's conjecture [1, Eq. (7a)] asks for nonnegative derivatives of every positive even order of this log-determinant objective in arbitrary dimension. The accompanying trace conjecture [1, Eq. (7b)] is a separate question. Li's fourth-order result [2] concerns the same log-determinant objective and is not limited to dimension two. Its proof uses fourth-order matrix-trace expansions and sign decompositions. Theorem 1 treats the sixth derivative in dimension two by comparing polynomial roots, including the noncommuting and singular cases permitted by (1).

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The determinant identity used below also appears in [2, Lemma 3]; the contribution here is its combination with ordered eigenvalue crossings to obtain the sixth-order sign on the full stated domain. The argument is self-contained and does not use either preprint's convexity results as premises. Low-dimensional matrix inequalities also occur in the Schatten-norm setting of Zeng, Liu, and Ratnavelu [3], cited here only as broad adjacent context.

The scope of the theorem is precisely dimension two, derivative order six, and interior weights. This paper does not resolve the arbitrary-dimension, all-even-order parent conjecture, the trace objective, endpoint derivative inequalities, or equality classification. A bounded primary-source screen on September 7, 2026 found no equivalent resolution of this particular case in the inspected sources; this is not an exhaustive priority claim.

2 Determinants and ordered crossings

Set $E = B + D$ and introduce the matrix polynomials

$$\begin{aligned} X(w) &= A + wB, & Y(w) &= C + (1 - w)D, \\ Z(w) &= (1 - w)X(w) + wY(w) = (1 - w)A + wC + w(1 - w)E. \end{aligned} \quad (4)$$

Write

$$a(w) = \det X(w), \quad b(w) = \det Y(w), \quad c(w) = \det Z(w). \quad (5)$$

On $(0, 1)$, $U \succeq A + B \succ 0$ and $V \succeq C + D \succ 0$. Consequently $X = wU$, $Y = (1 - w)V$, and $Z = w(1 - w)(U + V)$ are positive definite, so a, b, c are positive there. The multiplication identity

$$U^{-1}(U + V)V^{-1} = V^{-1} + U^{-1}$$

requires no commutation. Taking determinants and cancelling the scalar factors from the two-dimensional determinant scaling gives

$$\det P = \frac{\det U \det V}{\det(U + V)} = \frac{ab}{c}, \quad f := \log \det P = \log a + \log b - \log c. \quad (6)$$

We first record the crossing fact needed to locate these scalar roots.

Lemma 2. *Let $G(t)$ be a real symmetric 2×2 matrix function, real analytic in a neighborhood of every $t \geq 0$. Suppose $G(0) \succ 0$ and, for some $\beta > 0$,*

$$G(t) - G(s) \preceq -\beta(t - s)I_2 \quad (t > s \geq 0). \quad (7)$$

Its increasingly ordered eigenvalues $\lambda_1(t) \leq \lambda_2(t)$ each have exactly one positive zero, denoted $\tau_1 \leq \tau_2$. These are all the zeros of $\det G(t)$ on $[0, \infty)$, counted with algebraic multiplicity.

If K satisfies the same hypotheses, possibly with a different positive constant, and $G(t) \prec K(t)$ for every $t > 0$, then their indexed crossings satisfy $\tau_j(G) < \tau_j(K)$ for $j = 1, 2$.

Proof. The minimum and maximum of $x^\top G(t)x$ over unit vectors are $\lambda_1(t)$ and $\lambda_2(t)$. These formulas imply continuity: each eigenvalue changes by at most the operator norm of the matrix change. They also give

$$\lambda_j(t) \leq \lambda_j(s) - \beta(t - s) \quad (j = 1, 2).$$

Thus each eigenvalue decreases strictly from a positive initial value and tends to $-\infty$. Continuity gives exactly one positive zero for each; the eigenvalue ordering gives $\tau_1 \leq \tau_2$. The determinant vanishes exactly when an eigenvalue does.

It remains to verify the algebraic multiplicities without differentiating the ordered eigenvalues. From (7), $G'(\tau) \preceq -\beta I_2$ at any crossing τ . If $G(\tau)$ has nullity one, choose an orthonormal basis in which $G(\tau) = \text{diag}(0, d)$, $d \neq 0$. Its local expansion is

$$G(\tau + h) = \begin{pmatrix} h\alpha + O(h^2) & h\gamma + O(h^2) \\ h\gamma + O(h^2) & d + O(h) \end{pmatrix}, \quad \alpha < 0.$$

Hence $\det G(\tau + h) = h\alpha d + O(h^2)$, a simple zero. If the nullity is two, then $G(\tau) = 0$ and

$$\det G(\tau + h) = h^2 \det G'(\tau) + O(h^3).$$

Here $G'(\tau)$ is negative definite, so its determinant is positive; the zero has order two. This covers coincident crossings.

Finally, fix $t = \tau_j(K)$. Since $K(t) - G(t) \succ 0$, its smallest eigenvalue is some $\eta > 0$. The same minimum and maximum formulas give $\lambda_j(G(t)) \leq \lambda_j(K(t)) - \eta < 0$. The zero of $\lambda_j(G)$ therefore occurs strictly before t . $\square$

Proposition 3. *Assume temporarily that $A, B, C, D \succ 0$. Counting algebraic multiplicities, a has two roots $x_1 \geq x_2$ below zero, b has two roots $y_1 \leq y_2$ above one, and c has precisely four roots: $z_1 \geq z_2$ below zero and $r_1 \leq r_2$ above one. They satisfy*

$$x_j < z_j < 0 < 1 < r_j < y_j \quad (j = 1, 2). \quad (8)$$

Proof. For $t \geq 0$, define

$$H_L(t) = \frac{Z(-t)}{1+t} = A - tE - \frac{t}{1+t}C, \quad K_L(t) = A - tB, \quad (9)$$

$$H_R(t) = \frac{Z(1+t)}{1+t} = C - tE - \frac{t}{1+t}A, \quad K_R(t) = C - tD. \quad (10)$$

All four functions are real analytic on a neighborhood of $[0, \infty)$ and start positive definite. Moreover,

$$H'_L(t) = -E - \frac{C}{(1+t)^2}, \quad H'_R(t) = -E - \frac{A}{(1+t)^2}. \quad (11)$$

Both derivatives are bounded above by $-\lambda_{\min}(E)I_2$. The corresponding bounds for K'_L and K'_R use $\lambda_{\min}(B)$ and $\lambda_{\min}(D)$. Integrating these bounds verifies (7).

Let $\tau_1 \leq \tau_2$ be the crossings of H_L , and $\sigma_1 \leq \sigma_2$ those of K_L . For $t > 0$,

$$K_L(t) - H_L(t) = tD + \frac{t}{1+t}C \succ 0. \quad (12)$$

Lemma 2 gives $\tau_j < \sigma_j$. Since

$$a(-t) = \det K_L(t), \quad c(-t) = (1+t)^2 \det H_L(t),$$

the roots in question are $x_j = -\sigma_j$ and $z_j = -\tau_j$. The factor $(1+t)^2$ is nonzero for $t \geq 0$, and the affine substitution preserves multiplicities. Thus $x_j < z_j < 0$, with both lists in decreasing order as stated.

Likewise, let $\rho_1 \leq \rho_2$ be the crossings of H_R , and $\delta_1 \leq \delta_2$ those of K_R . The comparison

$$K_R(t) - H_R(t) = tB + \frac{t}{1+t}A \succ 0 \quad (t > 0) \quad (13)$$

gives $\rho_j < \delta_j$. Now

$$b(1+t) = \det K_R(t), \quad c(1+t) = (1+t)^2 \det H_R(t),$$

so $r_j = 1 + \rho_j$ and $y_j = 1 + \delta_j$, with $1 < r_j < y_j$.

Finally, a and b have degree two, with leading coefficients $\det B > 0$ and $\det D > 0$. The quadratic matrix polynomial Z has leading coefficient $-E$, so c has degree four and leading coefficient $\det E > 0$. The two negative and two greater-than-one roots just constructed exhaust this degree, with multiplicity. There are no remaining roots, real or nonreal. $\square$

3 The sixth derivative and singular inputs

Proof of Theorem 1. First suppose all four input matrices are positive definite. A real polynomial positive on $(0, 1)$, with all roots real and outside this interval, has logarithm equal to a constant plus the sum of $\log |w - s|$ over its roots s , counted with multiplicity. For each such root,

$$\frac{d^6}{dw^6} \log |w - s| = -\frac{120}{(w - s)^6}.$$

Applying Proposition 3 to (6) yields

$$\begin{aligned} \frac{f^{(6)}(w)}{120} &= \sum_{j=1}^2 \left(\frac{1}{(w - z_j)^6} - \frac{1}{(w - x_j)^6} \right) \\ &\quad + \sum_{j=1}^2 \left(\frac{1}{(w - r_j)^6} - \frac{1}{(w - y_j)^6} \right). \end{aligned} \quad (14)$$

For the first sum, $0 < w - z_j < w - x_j$; for the second, $0 < r_j - w < y_j - w$. The even exponent makes every difference in (14) strictly positive. Hence $f^{(6)}(w) > 0$ in this positive definite subcase.

Now assume only (1). For $\varepsilon > 0$, replace each input matrix $M \in \{A, B, C, D\}$ by $M_\varepsilon = M + \varepsilon I_2$, and use the subscript ε for the resulting polynomials and objective. The preceding argument gives $f_\varepsilon^{(6)}(w) > 0$.

Fix $w_0 \in (0, 1)$. Each of $a_\varepsilon, b_\varepsilon, c_\varepsilon$ is a polynomial in w whose coefficients are polynomial in ε . Its value at $(w, \varepsilon) = (w_0, 0)$ is positive by the determinant reduction. For any one of these polynomials write

$$q(w_0 + h) = \sum_{n \geq 0} q_n h^n, \quad \log q(w_0 + h) = \ell_0 + \sum_{n \geq 1} \ell_n h^n,$$

where $q_n = 0$ beyond the degree. Comparing the coefficient of h^{n-1} in $q(\log q)' = q'$ gives

$$\ell_n = \frac{nq_n - \sum_{k=1}^{n-1} k \ell_k q_{n-k}}{nq_0} \quad (n \geq 1). \quad (15)$$

For $n \leq 6$, this makes ℓ_n a rational function of $q_0, \dots, q_n$, with denominator a nonzero constant times a power of q_0 . Because q_0 remains positive near $\varepsilon = 0$, the sixth logarithmic Taylor coefficient is continuous in ε . Using (6) and $f^{(6)}(w_0) = 6![h^6]f(w_0 + h)$ therefore gives

$$f_\varepsilon^{(6)}(w_0) \longrightarrow f^{(6)}(w_0) \quad (\varepsilon \downarrow 0).$$

The positive values on the left have a nonnegative limit. Since w_0 was arbitrary, (3) follows on all of $(0, 1)$. $\square$

Remark 4. The fixed degrees and strict exterior-root locations in Proposition 3 are used only for positive definite inputs. They need not persist at a singular boundary. If $A = C = 0$ and $B = D = I_2$, then $c(w) = 4w^2(1 - w)^2$, with roots at the endpoints. If $B = D = 0$ and $A = C = I_2$, then $c(w) = 1$ and its degree drops to zero. In both examples the objective is constant. The continuity argument passes the derivative inequality, rather than a strict root-location assertion, to these boundaries.

4 Verification and assistance

The proof above is deductive and does not depend on computation. Exact rational checks were also performed with Python 3.9.6 and SymPy 1.14.0: root-isolation certificates for 15 positive definite quadruples, and 60 sixth-derivative comparisons at $w \in \{1/1000, 1/2, 999/1000\}$ for those quadruples and five semidefinite boundary cases. The root tests include repeated roots and 13 quadruples with a noncommuting pair. The derivative values are all nonnegative, including six exact zeros. Four general symbolic identities check the two substitutions (9)–(10) and the two derivative formulas (11).

A separate verification implementation reconstructed all 15 root certificates using exact Euclidean and Sturm calculations, and all 60 derivative values from direct symbolic inversion of (2) and Newton power sums. It also checked a noncommuting positive definite example with two double exterior roots. These are finite corroborations, not an exhaustive search over the matrix domain. The accompanying verification materials describe the inputs and exact arithmetic scripts.

OpenAI Codex assisted with the proof, verification programs, and manuscript. A separate model context reviewed the proof and implemented the second verifier. This review is neither journal peer review nor formal proof-assistant certification. The explicit argument above is the basis for the mathematical claim.

References

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